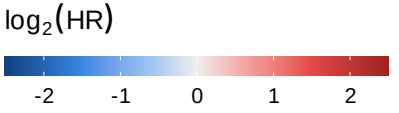
B Post-progression death by first-progression pattern (reference: IH-NL)

All-cause: Cox HR; cause-specific: csHR (cause-specific Cox). Lunn–McNeil heterogeneity across IH/EH/Systemic HCC death: pattern × cause $\chi^2(16) = 60.2$, $P < 0.001$; site × cause $\chi^2(4) = 48.8$, $P < 0.001$; form × cause $\chi^2(4) = 7.7$, $P = 0.104$

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.17 (0.98–1.40)	1.10 (0.89–1.38)	1.68 (0.87–3.26)	1.16 (0.74–1.82)	1.02 (0.53–1.94)
IH-NL	Reference	Reference	Reference	Reference	Reference
IH-TG+NL	1.58 (1.23–2.05)***	1.66 (1.22–2.25)**	2.53 (1.01–6.30)*	1.40 (0.73–2.69)	1.00 (0.33–3.00)
EH-TG	1.83 (1.41–2.39)***	1.57 (1.11–2.24)*	4.46 (2.13–9.34)***	2.01 (1.12–3.61)*	1.09 (0.35–3.32)
EH-NL	1.26 (1.03–1.53)*	0.84 (0.64–1.10)	3.01 (1.60–5.66)***	1.95 (1.27–3.01)**	1.68 (0.88–3.20)
EH-TG+NL	1.72 (1.10–2.68)*	1.14 (0.59–2.19)	4.10 (1.12–14.96)*	2.32 (0.99–5.46)	4.10 (1.14–14.73)*
Sys-TG	2.02 (1.42–2.87)***	1.55 (0.95–2.54)	1.91 (0.53–6.85)	4.12 (2.22–7.64)***	NE (1 event)
Sys-NL	1.70 (1.28–2.25)***	1.03 (0.68–1.56)	5.86 (2.73–12.59)***	3.30 (1.92–5.68)***	NE (2 events)
Sys-TG+NL	1.78 (1.44–2.20)***	1.27 (0.97–1.68)	3.22 (1.57–6.60)**	3.64 (2.34–5.64)***	1.07 (0.41–2.77)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.405**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group. NE: <3 events in the cell. * P<0.05, ** P<0.01, *** P<0.001 (unadjusted; BH-FDR q values in Table S-HR). Overall Wald test of the 8 pattern terms does not depend on the reference.